YEAR 10 SCIENCE LESSON 10

Can electricity make a magnet?

Names: _____

1 Build it

Wind the wire around the nail. Count your turns. Ask your teacher to check it before you connect the power.

How many turns? _____

2 Try it

How many clips can the nail lift with the power off? _____

How many with the power on? _____ Switch off again.

Talk with your partner. Why do you think that happened?

3 Change the turns

Switch off and disconnect before changing the coil. Use the same nail, wire, current and clips each time. Test each coil twice.

Fewer turns: _____ Clips lifted: _____ and _____

More turns: _____ Clips lifted: _____ and _____

Which lifted more? _____

Talk with your partner. What do you think made the difference?